

43.5 Choosing the smoothing parameter

Four smoothers have appeared: the local constant and local linear estimators of [Section 43.1](#), the penalized spline of [Section 43.3](#) and the smoothing spline of [Section 43.4](#). Each is linear in $\mathbf{y}$, each has one dial, and none has said how to set it. The rules of this section approximate a risk that cannot be observed: they agree when the sample is large and the errors independent, and fail in recognisably different ways when it is not.

43.5.1 Every smoother here is a linear operator

Proposition 43.5.1. Linear smoothers

Call $\hat{\mathbf{f}} = \mathbf{S}\mathbf{y}$ a **linear smoother** with **smoother matrix** $\mathbf{S}$ when $\mathbf{S}$ does not depend on $\mathbf{y}$. The estimators of Definition 43.1.1, Definition 43.3.1 and Theorem 43.4.2 are linear smoothers, with $\mathbf{S}$ given by Eq. 43.1.4, Eq. 43.3.2 and Eq. 43.4.2. Suppose $E(\mathbf{Y}) = \mathbf{f}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Then:

- (a) (*Moments.*) $E(\hat{\mathbf{f}}) = \mathbf{S}\mathbf{f}$, $\text{Cov}(\hat{\mathbf{f}}) = \sigma^2 \mathbf{S}\mathbf{S}^\top$, and

$$E\|\hat{\mathbf{f}} - \mathbf{f}\|^2 = \|(\mathbf{I} - \mathbf{S})\mathbf{f}\|^2 + \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top). \quad (43.5.1)$$

- (b) (*Residuals.*) $E\|\mathbf{y} - \hat{\mathbf{f}}\|^2 = \|(\mathbf{I} - \mathbf{S})\mathbf{f}\|^2 + \sigma^2\{n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}\mathbf{S}^\top)\}$, so

$$\hat{\sigma}^2 = \frac{\|\mathbf{y} - \hat{\mathbf{f}}\|^2}{n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}\mathbf{S}^\top)} \quad (43.5.2)$$

is unbiased for σ^2 when the bias term is negligible.

- (c) (*Degrees of freedom.*) If $\mathbf{S}$ is symmetric with eigenvalues in $[0, 1]$ — as it is for a penalized or smoothing spline, by Proposition 43.3.2(b) — then $\text{tr}(\mathbf{S}\mathbf{S}^\top) \leq \text{tr}(\mathbf{S}) \leq 2\text{tr}(\mathbf{S}) - \text{tr}(\mathbf{S}\mathbf{S}^\top)$, with equality throughout iff $\mathbf{S}$ is an orthogonal projection, in which case all three equal its rank.
- (d) (*Shrinkage.*) For a penalized spline, Eq. 43.3.2 writes $\mathbf{S}_\lambda$ in the Demmler–Reinsch basis as a diagonal shrinkage $\mathbf{u}_j^\top \mathbf{y} \mapsto \mathbf{u}_j^\top \mathbf{y} / (1 + \lambda e_j)$: smooth directions (e_j small) pass through, rough ones are damped. The smoothing spline has the same form with e_j the eigenvalues of the roughness matrix relative to the identity.
- (e) (*Pointwise bands.*) $\hat{f}(x_i) \pm z_{1-\alpha/2} \hat{\sigma} \sqrt{(\mathbf{S}\mathbf{S}^\top)_{ii}}$ is an interval for $E\{\hat{f}(x_i)\} = (\mathbf{S}\mathbf{f})_i$, not for $f(x_i)$. Its coverage of $f(x_i)$ is $1 - \alpha$ only to the extent that the bias $\{(\mathbf{S} - \mathbf{I})\mathbf{f}\}_i$ is small next to $\sigma \sqrt{(\mathbf{S}\mathbf{S}^\top)_{ii}}$.

Proof. Linearity is Proposition 43.1.1 for the kernel estimators, Proposition 43.3.2(a) for the penalized spline and Eq. 43.4.2 for the smoothing spline; in each case the matrix is built from the design, the knots and the tuning constant alone.

- (a) $E(\mathbf{S}\mathbf{Y}) = \mathbf{S}\mathbf{f}$ and $\text{Cov}(\mathbf{S}\mathbf{Y}) = \mathbf{S}\text{Cov}(\mathbf{Y})\mathbf{S}^\top$. For the risk, $\hat{\mathbf{f}} - \mathbf{f} = (\mathbf{S} - \mathbf{I})\mathbf{f} + \mathbf{S}\boldsymbol{\varepsilon}$, the two terms are uncorrelated, and $E\|\mathbf{S}\boldsymbol{\varepsilon}\|^2 = \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$.
- (b) $\mathbf{y} - \hat{\mathbf{f}} = (\mathbf{I} - \mathbf{S})\mathbf{f} + (\mathbf{I} - \mathbf{S})\boldsymbol{\varepsilon}$, and

$$\begin{aligned} E\|(\mathbf{I} - \mathbf{S})\boldsymbol{\varepsilon}\|^2 &= \sigma^2 \text{tr}\{(\mathbf{I} - \mathbf{S})(\mathbf{I} - \mathbf{S})^\top\} \\ &= \sigma^2\{n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}\mathbf{S}^\top)\}. \end{aligned}$$

- (c) With eigenvalues $s_j \in [0, 1]$ we have $\text{tr}(\mathbf{S}) = \sum_j s_j$ and $\text{tr}(\mathbf{S}\mathbf{S}^\top) = \sum_j s_j^2$, while $s_j^2 \leq s_j \leq 2s_j - s_j^2$, the last inequality because $(1 - s_j)^2 \geq 0$. Equality in either needs every $s_j \in \{0, 1\}$, which for a symmetric matrix means $\mathbf{S}$ is idempotent (Theorem 6.3.3); its trace is then its rank (Proposition 6.3.4).
- (d) is Eq. 43.3.2; for the smoothing spline apply Lemma 43.3.1 with $\mathbf{B} = \mathbf{I}$ and the roughness matrix of Proposition 43.4.3(b) in place of $\mathbf{P}$.
- (e) By (a), $\hat{f}(x_i)$ has mean $(\mathbf{S}\mathbf{f})_i$ and standard deviation $\sigma\{(\mathbf{S}\mathbf{S}^\top)_{ii}\}^{1/2}$, so under normal errors the interval covers that mean with probability $1 - \alpha$ if $\hat{\sigma}$ is replaced by σ . Writing $f(x_i) = (\mathbf{S}\mathbf{f})_i - \{(\mathbf{S} - \mathbf{I})\mathbf{f}\}_i$ shows that its coverage of $f(x_i)$ is the normal probability of an interval displaced by the bias.

■

Warning. Part (e) is the most important caveat in this chapter, and the one most often ignored. A smoother is *deliberately* biased; that is what the penalty buys. The interval above treats the bias as zero, so where f curves most — a peak, a trough, the boundary — it is centred in the wrong place and undercovers, while on flat stretches it overcovers. [Section 43.6](#) gives a band that repairs the average of this, and says what “average” means.

43.5.2 Three rules

The quantity one would like to minimize is the risk [Eq. 43.5.1](#), which involves the unknown $\mathbf{f}$. Each rule estimates something close to it.

Theorem 43.5.2. Cross-validation, generalized cross-validation and AIC for smoothers

Let $\mathbf{S}_\lambda$ be the smoother matrix, $\hat{\mathbf{f}} = \mathbf{S}_\lambda \mathbf{y}$, $\text{df} = \text{tr}(\mathbf{S}_\lambda)$ and s_{ii} the diagonal entries of $\mathbf{S}_\lambda$ — the leverage of Proposition 6.8.1 and [Section 20.2](#), read now for a smoother instead of a projection.

- (a) (*The leave-one-out shortcut is exact.*) For a penalized spline and for a smoothing spline,

$$\text{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^n \{y_i - \hat{f}^{(-i)}(x_i)\}^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}(x_i)}{1 - s_{ii}} \right)^2, \quad (43.5.3)$$

where $\hat{f}^{(-i)}$ is the fit computed without case i at the same λ . The same identity holds for a local polynomial estimator, with $s_{ii} = \ell_i(x_i)$.

- (b) (*Generalized cross-validation.*) Replacing each s_{ii} by the average df/n gives Definition 29.3.2,

$$\text{GCV}(\lambda) = \frac{n^{-1} \|\mathbf{y} - \hat{\mathbf{f}}\|^2}{(1 - \text{df}/n)^2}, \quad (43.5.4)$$

which needs only the trace, and satisfies $\text{GCV} = n^{-1} \|\mathbf{y} - \hat{\mathbf{f}}\|^2 (1 + 2 \text{df}/n) + O(\text{df}^2/n^2)$.

- (c) (*Information criteria.*) Under normal errors with σ^2 profiled out, Proposition 29.2.4 gives $\text{AIC}(\lambda) = n \log(\|\mathbf{y} - \hat{\mathbf{f}}\|^2/n) + 2 \text{df}$. Its small-sample correction (Hurvich, Simonoff and Tsai, 1998) is

$$\text{AIC}_c(\lambda) = \log\left(\frac{\|\mathbf{y} - \hat{\mathbf{f}}\|^2}{n}\right) + 1 + \frac{2(\text{df} + 1)}{n - \text{df} - 2}, \quad (43.5.5)$$

which penalizes large df much more heavily and has a pole at $\text{df} = n - 2$.

- (d) (*What they estimate.*) If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $E\{\text{CV}(\lambda)\}$ is, up to $O(n^{-1})$ terms, the average prediction error at a fresh response, which exceeds n^{-1} times the risk [Eq. 43.5.1](#) by σ^2 . A minimizer of any of the three is therefore aiming at [Eq. 43.5.1](#), not at λ itself, and nothing here estimates a “true” smoothing parameter, since no such thing is assumed to exist.

Proof.

- (a) A penalized spline minimizes $\|\mathbf{y} - \mathbf{B}\boldsymbol{\gamma}\|^2 + \boldsymbol{\gamma}^\top \boldsymbol{\Omega} \boldsymbol{\gamma}$ with $\boldsymbol{\Omega} = \lambda \mathbf{P}$, and a smoothing spline minimizes $\|\mathbf{y} - \mathbf{z}\|^2 + \mathbf{z}^\top (\lambda \mathbf{P}) \mathbf{z}$, which is the same form with $\mathbf{B} = \mathbf{I}$. Both have $\boldsymbol{\Omega}$ nonnegative definite and the deleted cross-product matrices nonsingular, so Theorem 29.3.2 applies verbatim and gives [Eq. 43.5.3](#). For a local polynomial estimator, fix i and let $\tilde{\mathbf{y}}$ be $\mathbf{y}$ with y_i replaced by $\hat{f}^{(-i)}(x_i)$. The weighted least squares problem [Eq. 43.1.2](#) at $x = x_i$ with data $\tilde{\mathbf{y}}$ has the same minimizer as the problem with case i deleted. Indeed the row of $\mathbf{U}_x$ belonging to case i is $(1, 0, \dots, 0)$, so the objective with $\tilde{\mathbf{y}}$ is the deleted objective plus $K_h(0)(\tilde{y}_i - b_0)^2$; the deleted minimizer makes the first term as small as possible and the second term zero, so it minimizes the sum. Hence $\hat{f}^{(-i)}(x_i) = \sum_j \ell_j(x_i) \tilde{y}_j = \hat{f}(x_i) + \ell_i(x_i) \{\hat{f}^{(-i)}(x_i) - y_i\}$; solving for $y_i - \hat{f}^{(-i)}(x_i)$ gives $\{y_i - \hat{f}(x_i)\} / \{1 - \ell_i(x_i)\}$.
- (b) The substitution is the definition of GCV; the expansion is $(1 - t)^{-2} = 1 + 2t + O(t^2)$ with $t = \text{df}/n$.
- (c) With a normal likelihood and σ^2 replaced by its maximum likelihood estimate $\|\mathbf{y} - \hat{\mathbf{f}}\|^2/n$, the maximized log likelihood is $-\frac{n}{2} \log(\|\mathbf{y} - \hat{\mathbf{f}}\|^2/n) + \text{const}$, and Proposition 29.2.4 counts $\text{df} + 1$ parameters, the $+1$ being σ^2 ; dropping constants gives AIC. [Eq. 43.5.5](#) is the stated correction, which reduces to AIC/n plus constants when $\text{df}/n \rightarrow 0$.
- (d) is Proposition 29.3.1 and Definition 29.1.1, which do not use the form of the estimator. ■

Idea. All three rules do the same arithmetic: residual sum of squares, inflated by a factor that grows with the degrees of freedom — $(1 - s_{ii})^{-2}$ case by case, $(1 - \text{df}/n)^{-2}$ on average, $\exp\{2 \text{df}/n\}$ approximately for AIC. That is why they usually agree, and why they part company at large df. Mallows' C_p (Definition 29.2.1) inflates additively, $\|\mathbf{y} - \hat{\mathbf{f}}\|^2 + 2\hat{\sigma}^2 \text{tr}(\mathbf{S}_\lambda)$, but needs a $\hat{\sigma}^2$ from outside — usually [Eq. 43.5.2](#) at a deliberately undersmoothed fit — and inherits whatever that choice got wrong.

43.5.3 What they do in practice

Example. The three criteria on one data set, and over 200 of them

Take $n = 120$ equally spaced design points, the mean function of Example (Bias, variance

and the optimal bandwidth), $\sigma = 0.25$, and a cubic P-spline with $K = 25$ knots and a second-order penalty. On one data set the shortcut [Eq. 43.5.3](#) agrees with 120 actual refits to 4.6×10^{-15} , and the three criteria choose 9.83, 10.29 and 9.40 degrees of freedom.

Over 200 replicates the three are hard to tell apart. Each has median selected df 7.82, and the average $n^{-1}\|\hat{\mathbf{f}} - \mathbf{f}\|^2$ is 0.00448 for CV, 0.00449 for GCV and 0.00429 for AIC_c . The differences show in the upper tail: CV and GCV choose more than 15 degrees of freedom in 2.5 per cent of the samples, AIC_c in 0.5 per cent. The criteria are flat to the right of the minimum ([Figure 43.5.1\(a\)](#)), so an occasional sample sends the minimum far out, and steep to the left, so gross oversmoothing is rare.

Warning (Three small-sample failures). **A flat criterion.** Near the minimum, GCV can change by a fraction of a per cent over a factor of two in λ ; the choice is then arbitrary, and reporting the fit at a single λ hides that. Plot the criterion.

Interpolation. Since $(1 - \text{df}/n)^{-2}$ becomes large only as $\text{df} \rightarrow n$, a small sample can give GCV a *local* minimum at the interpolating end. AIC_c cannot, because of the pole in [Eq. 43.5.5](#); this is its main practical advantage.

Correlated errors. All three estimate prediction error at a *new, independent* response. If the errors are serially correlated, a wiggly fit predicts its neighbours well without describing f at all, and every criterion rewards it ([Example \(What correlated errors do\)](#)).

Example. What correlated errors do

Repeat the simulation with errors from a stationary AR(1) process of the same marginal standard deviation and autocorrelation $\rho = 0.7$ (Definition 21.6.1). The median selected degrees of freedom rise from 7.82 to 25.40 for CV, 24.53 for GCV and 23.50 for AIC_c , and the average squared error rises from the values just quoted to 0.04209, 0.04145 and 0.03982 — roughly a ninefold loss. The fitted curve in [Figure 43.5.1\(b\)](#) uses 27 degrees of freedom and traces every excursion of the error process.

There is no repair inside this section: a smooth mean and smooth-looking noise are not separately identified without an assumption about one of them. The honest responses are to model the correlation (Part VII), to choose λ by a criterion that accounts for it — leaving out whole blocks rather than single points, or REML in the mixed-model form of [Section 43.6](#) — or to say that the wiggles are not interpretable. Opsomer, Wang and Yang (2001) survey what is known.

```
import numpy as np

def f_true(x):
    """The mean function of Section 43.1."""
    return 2 * x + np.exp(-25 * (x - 0.35) ** 2) - 0.75 * np.exp(-50 * (x - 0.80) ** 2)

def uniform_knots(a, b, K, d):
    """K interior knots equally spaced in (a, b), continued d further steps at each end."""
    delta = (b - a) / (K + 1)
```

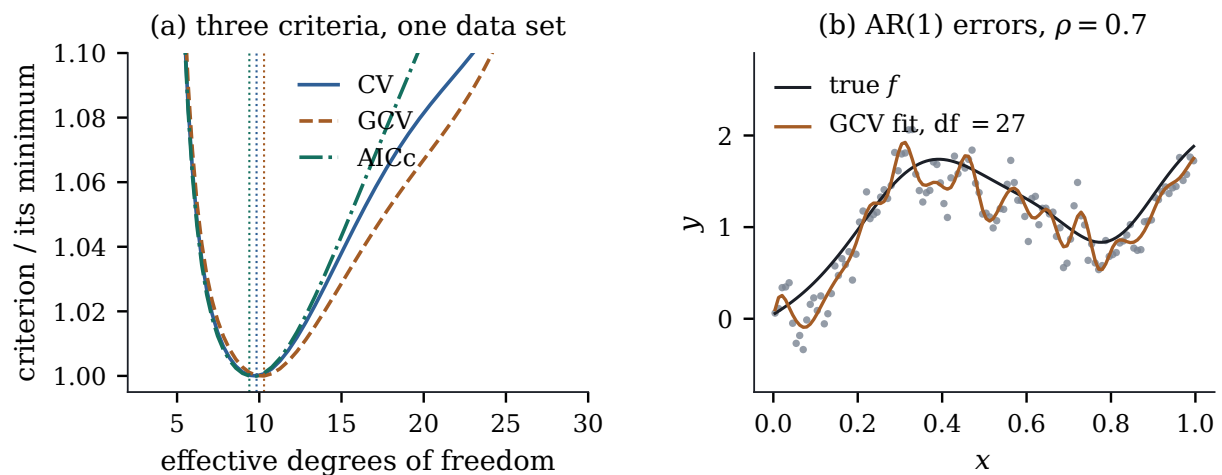


Figure 43.5.1: (a) The three criteria on one data set, each divided by its own minimum and plotted against effective degrees of freedom; dotted lines mark the three choices. All are steep on the left and flat on the right. (b) One data set with AR(1) errors, $\rho = 0.7$, and the fit chosen by GCV.

```

return a + delta * np.arange(-d, K + 2 + d)

def bspline_basis(x, kn, d):
    """The B-splines of degree d on the knot vector kn, by the de Boor recurrence.
    """
    x = np.atleast_1d(np.asarray(x, float))
    B = np.array([(x >= kn[j]) & (x < kn[j + 1]) for j in range(len(kn) - 1)], float)
    B.T
    last = np.max(np.nonzero(kn[:-1] < kn[1:])[0])
    B[x == kn[-1], last] = 1.0
    for deg in range(1, d + 1):
        C = np.zeros((len(x), B.shape[1] - 1))
        for j in range(C.shape[1]):
            if kn[j + deg] > kn[j]:
                C[:, j] += (x - kn[j]) / (kn[j + deg] - kn[j]) * B[:, j]
            if kn[j + deg + 1] > kn[j + 1]:
                C[:, j] += (kn[j + deg + 1] - x) / (kn[j + deg + 1] - kn[j + 1]) * B
       [:, j + 1]
        B = C
    return B

def smoother(B, P, lam):
    """The smoother matrix of the penalized fit with basis B and penalty matrix P.
    """
    return B @ np.linalg.solve(B.T @ B + lam * P, B.T)

def criteria(B, P, y, lam):
    """Leave-one-out CV by the leverage shortcut, GCV, and the corrected AIC."""
    n = len(y)
    S = smoother(B, P, lam)
    resid = y - S @ y
    df = np.trace(S)
    cv = np.mean((resid / (1 - np.diag(S))) ** 2)

```

```

gcv = np.mean(resid**2) / (1 - df / n) ** 2
aicc = np.log(np.mean(resid**2)) + 1 + 2 * (df + 1) / (n - df - 2)
return df, cv, gcv, aicc

n, sigma, K, d, k = 120, 0.25, 25, 3, 2
x = (np.arange(1, n + 1) - 0.5) / n
kn = uniform_knots(0.0, 1.0, K, d)
B = bspline_basis(x, kn, d)
Dk = np.diff(np.eye(B.shape[1]), n=k, axis=0)
P = Dk.T @ Dk
rng = np.random.default_rng(4343)
y = f_true(x) + sigma * rng.normal(size=n)

lams = np.geomspace(1e-9, 1e3, 121)
table = np.array([criteria(B, P, y, lam) for lam in lams])
dfs, cvs, gcvs, aiccs = table.T
pick = {"CV": lams[np.argmin(cvs)], "GCV": lams[np.argmin(gcvs)], "AICc": lams[
    np.argmin(aiccs)]}
df_pick = {name: criteria(B, P, y, lam)[0] for name, lam in pick.items()}
print("degrees of freedom chosen:", {k_: round(v, 2) for k_, v in df_pick.items()})

```

[Run this code online](#)

43.5.4 After the choice

Remark (Inference after selecting the smoothing parameter). Every interval, test and standard error in this chapter is computed *as if* λ were fixed in advance. It was not: it was chosen by minimizing a criterion computed from the same $\mathbf{y}$. This is Proposition 29.5.1 in a continuous guise. The residual sum of squares at the selected λ is the smallest of many, so $\hat{\sigma}^2$ from Eq. 43.5.2 is biased downwards and a pointwise band does not attain its nominal coverage.

The effect is modest when the criterion has a well-determined minimum, which is why the practice survives, and large when it does not. The defences are those of Chapter 29: hold out data, bootstrap the whole procedure including the selection of λ (Chapter 23), or report the fit at several λ and let the reader see the dependence. What is not a defence is quoting a p -value from a fit whose flexibility was chosen to make that p -value small.

43.5.5 Exercises

43.5.5.1 A. Check your understanding

Exercise. A1

For the projection onto a p -dimensional space, compute $\text{tr}(\mathbf{S})$, $\text{tr}(\mathbf{S}\mathbf{S}^\top)$ and $2\text{tr}(\mathbf{S}) - \text{tr}(\mathbf{S}\mathbf{S}^\top)$. For a P-spline at $\text{df} = 8$ with $m = 24$, which of the three is largest? Using Proposition 43.5.1(c), say which one the residual degrees of freedom $n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}\mathbf{S}^\top)$ subtracts.

43.5.5.2 B. Practice

Exercise. B1

Using the cell above, verify [Eq. 43.5.3](#) against n refits at three values of λ , including one so large that $\text{df} \approx 2$ and one so small that $\text{df} \approx m$. Then do the same for a *local linear* estimator, using $\ell_i(x_i)$ from [Section 43.1](#).

Exercise. B2

Show that $\log \text{GCV} = \log(n^{-1}\|\mathbf{y} - \hat{\mathbf{f}}\|^2) - 2\log(1 - \text{df}/n)$ and that $-2\log(1 - t) = 2t + t^2 + O(t^3)$. Compare with [Eq. 43.5.5](#) expanded for small df/n , and explain why GCV and AIC almost always choose the same λ when n is large and differ when df is an appreciable fraction of n .

Solution. The first identity is immediate from [Eq. 43.5.4](#). Expanding, $\log \text{GCV} = \log(\text{RSS}/n) + 2\text{df}/n + (\text{df}/n)^2 + \dots$, while $\text{AIC}_c = \log(\text{RSS}/n) + 1 + 2(\text{df} + 1)/(n - \text{df} - 2) = \log(\text{RSS}/n) + 1 + 2\text{df}/n + 2\text{df}^2/n^2 + \dots$. To first order in df/n the two differ by a constant, so their minimizers agree; the second-order terms differ by a factor of two, and AIC_c has a pole at $\text{df} = n - 2$ where GCV has one only at $\text{df} = n$. Hence they separate exactly when df/n is not small.

43.5.5.3 C. Going deeper**Exercise. C1**

GCV was introduced by Craven and Wahba (1979) as a version of cross-validation invariant to rotation. Show that if $\mathbf{\Gamma}$ is orthogonal and one replaces $\mathbf{y}$ by $\mathbf{\Gamma y}$ and $\mathbf{S}_\lambda$ by $\mathbf{\Gamma S}_\lambda \mathbf{\Gamma}^\top$, then GCV is unchanged while CV is not. Why is this a virtue for ridge regression (where such rotations are natural) and less obviously one for a smoother (where they are not)?

Solution. Under the substitution, the residual vector becomes $\mathbf{\Gamma}(\mathbf{I} - \mathbf{S}_\lambda)\mathbf{y}$, whose squared length is unchanged, and $\text{tr}(\mathbf{\Gamma S}_\lambda \mathbf{\Gamma}^\top) = \text{tr}(\mathbf{S}_\lambda)$; so [Eq. 43.5.4](#) is invariant, while CV depends on the individual s_{ii} , which are not. In ridge regression the canonical coordinates of [Theorem 27.2.2](#) are as natural as the original ones, so invariance is attractive; for a smoother the coordinates are cases in a fixed order along the x -axis and the s_{ii} carry real information — small in dense regions, large in sparse ones — which replacing them by their average throws away. Hence CV and GCV differ most in designs with isolated points.

Exercise. C2

For the simulation of [Example \(The three criteria on one data set, and over 200 of them\)](#), record for each replicate the *set* of λ within one per cent of the GCV minimum, and report the median ratio of the largest to the smallest df in that set. Then plot, for one replicate, the fits at both ends of the interval. Does the choice within the flat region matter for what one would say about f ?